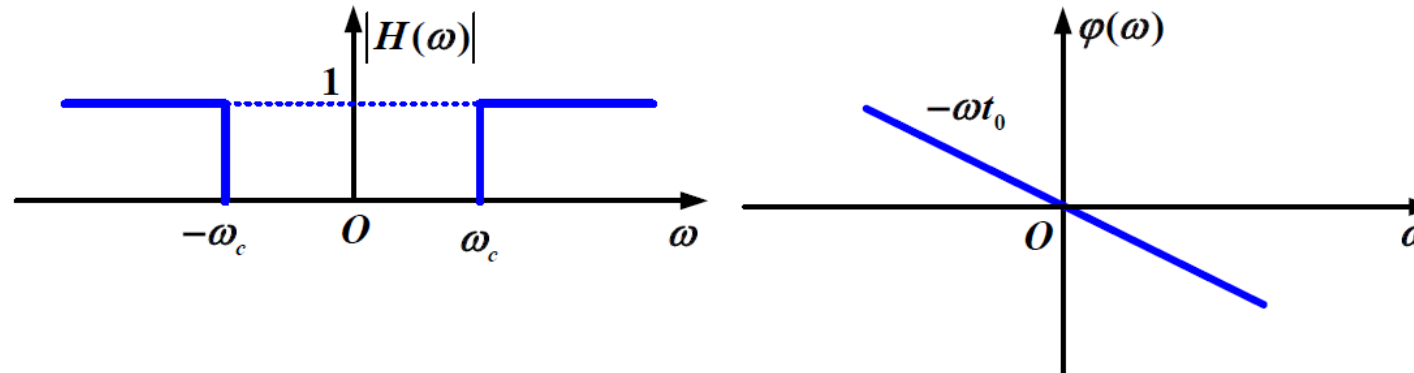


1 Ideal Low Pass Filter

Example 1

The figure below shows the magnitude-frequency and phase-frequency characteristics of an **ideal high-pass filter**

Determine the **impulse response** of this ideal high-pass filter



$$|H(\omega)| = 1 - [u(\omega + \omega_c) - u(\omega - \omega_c)]$$

$$\phi(\omega) = -\omega t_0$$

$$h(t) = \delta(t - t_0) - \frac{\omega_c}{\pi} \text{Sa}[\omega_c(t - t_0)]$$

1 Ideal Low Pass Filter

How about

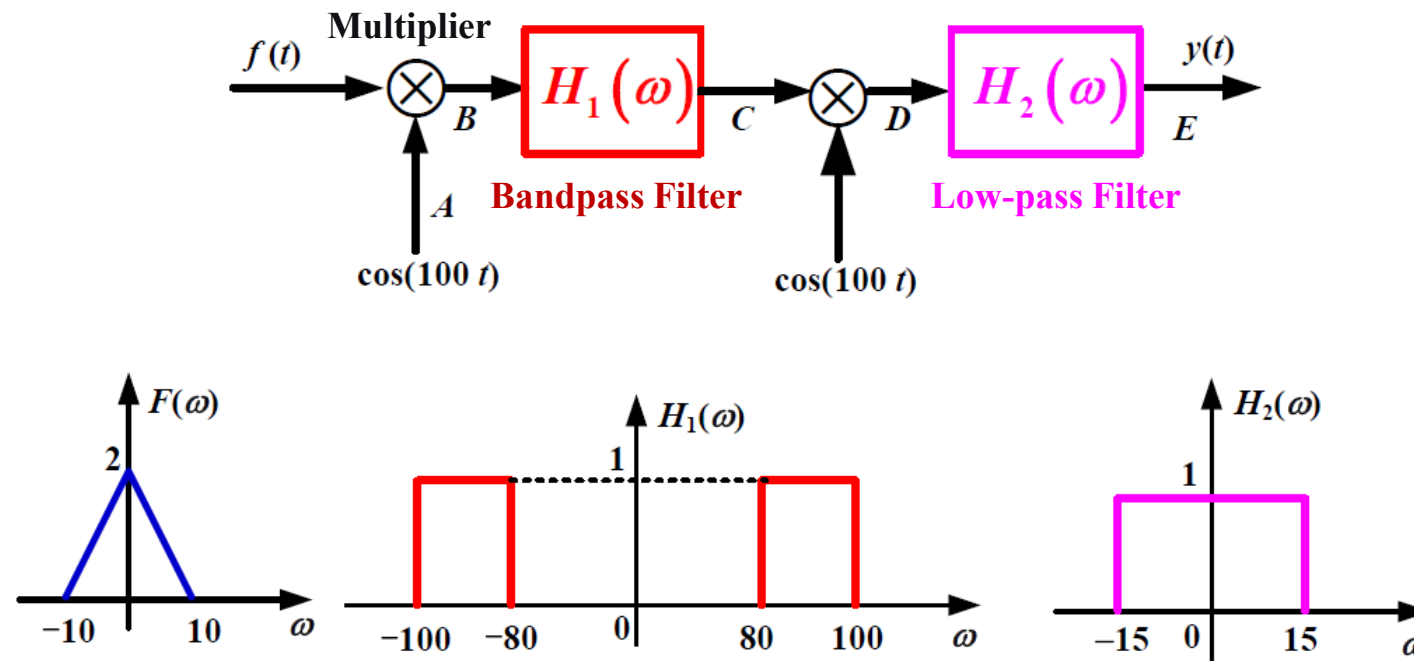
Ideal Low Pass Filter?

Ideal Bandpass Filter?

2 Modulation and demodulation

Example 2

In the system shown in the figure, given the spectrum $F(\omega)$ of the input signal $f(t)$, analyze the spectra at points A , B , C , D , and E in the system and sketch the spectral diagrams. Determine the relationship between $y(t)$ and $f(t)$

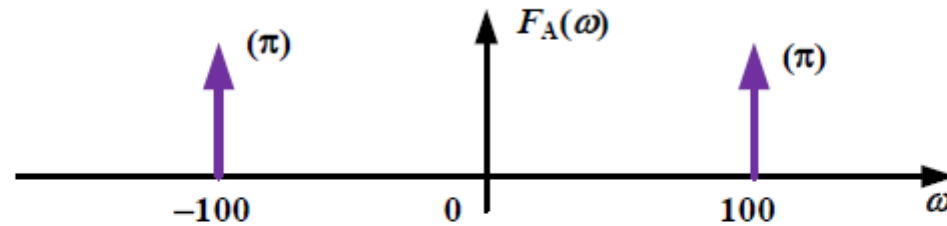


2 Modulation and demodulation



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$$F_A(\omega) = \mathcal{F}[\cos(100t)] = \pi[\delta(\omega - 100) + \delta(\omega + 100)]$$



$$f_B(t) = f(t) \cos(100t)$$

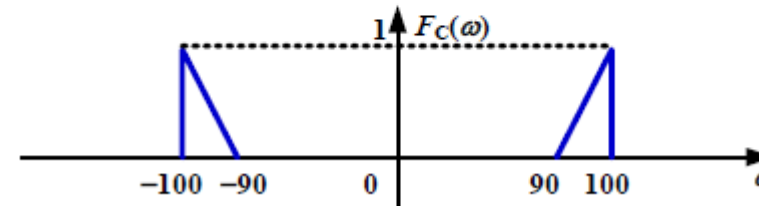
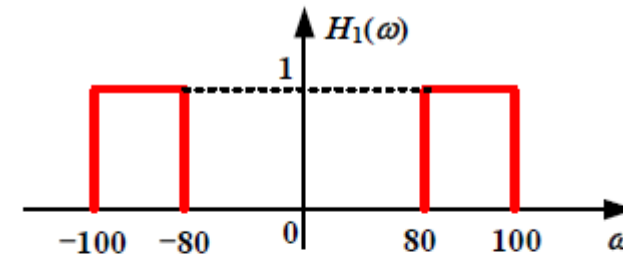
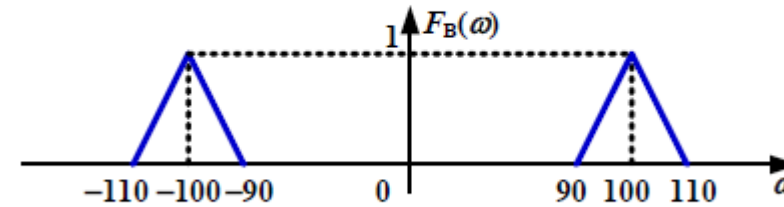
$$F_B(\omega) = \frac{1}{2}[F(\omega - 100) + F(\omega + 100)]$$

2 Modulation and demodulation



$$F_B(\omega) = \frac{1}{2} [F(\omega - 100) + F(\omega + 100)]$$

$$F_C(\omega) = F_B(\omega) H_1(\omega)$$



2 Modulation and demodulation

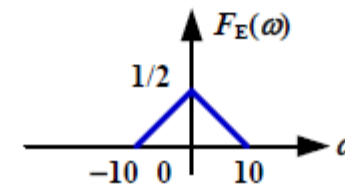
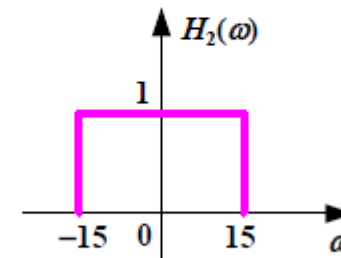
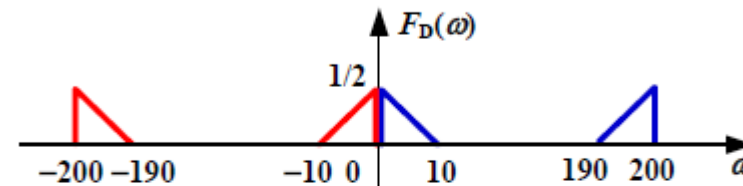
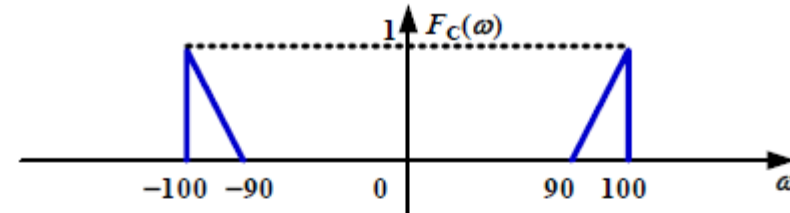


$$f_D(t) = f_C(t) \cos(100t)$$

$$F_D(\omega) = \frac{1}{2} [F_C(\omega - 100) + F_C(\omega + 100)]$$

$$F_E(\omega) = Y(\omega) \\ = F_D(\omega) H_2(\omega)$$

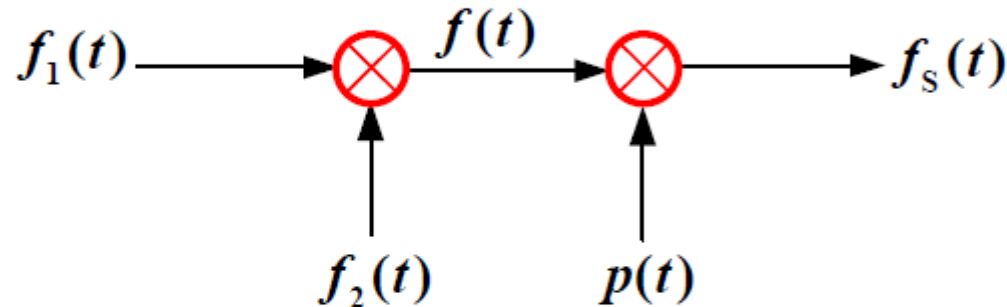
$$y(t) = \frac{1}{4} f(t)$$



3 Sampling and the Sampling Theorem

Example 3

The system is shown in the figure



$$f_1(t) = \text{Sa}(1000\pi t)$$

$$f_2(t) = \text{Sa}(2000\pi t)$$

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

- (1) To recover $f(t)$ from $f_s(t)$ without distortion, find the maximum sampling interval $T_{\max}$
- (2) When $T = T_{\max}$, sketch the spectrum $F_s(\omega)$ of $f_s(t)$

3 Sampling and the Sampling Theorem



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$$(1) \quad f_1(t) = \text{Sa}(\mathbf{1000\pi}t)$$

$$F_1(\omega) = \frac{\pi}{\omega_1} \left[u(\omega + \omega_1) - u(\omega - \omega_1) \right] \quad (\omega_1 = \mathbf{1000\pi})$$

$$f_2(t) = \text{Sa}(\mathbf{2000\pi}t)$$

$$F_2(\omega) = \frac{\pi}{\omega_2} \left[u(\omega + \omega_2) - u(\omega - \omega_2) \right] \quad (\omega_2 = \mathbf{2000\pi})$$

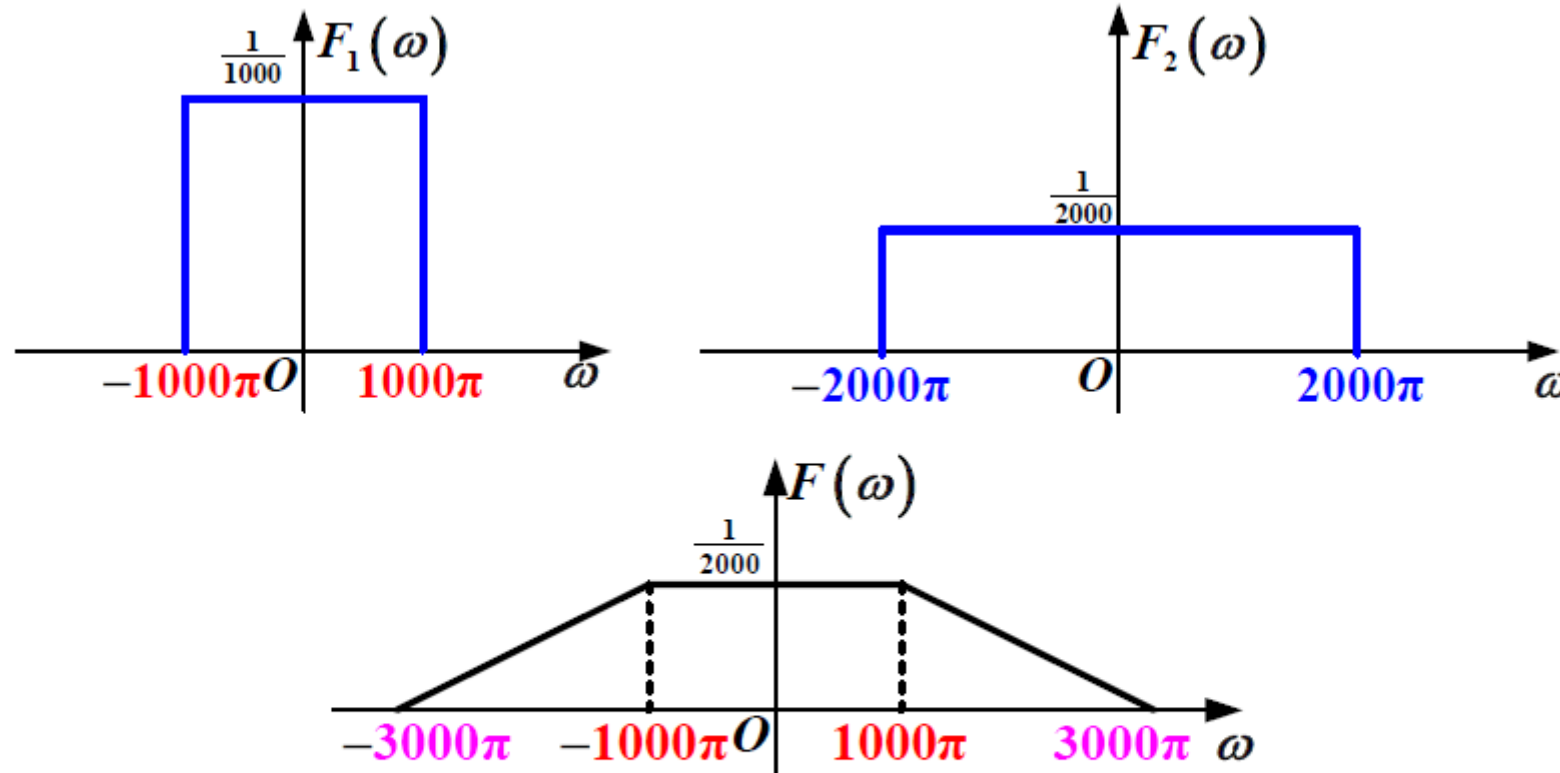
$$f(t) = f_1(t)f_2(t)$$

$$\therefore F(\omega) = \frac{\mathbf{1}}{\mathbf{2\pi}} F_1(\omega) * \mathbf{F_2}(\omega)$$

3 Sampling and the Sampling Theorem



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$$\therefore \omega_m = \omega_1 + \omega_2 = 3000\pi \text{ rad/s} \quad T_{\max} = \frac{2\pi}{2\omega_m} = \frac{1}{3000} \text{ s}$$

3 Sampling and the Sampling Theorem



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(2)

$$\therefore f_s(t) = f(t) \cdot p(t)$$

when $\omega_s = \frac{2\pi}{T_{\max}}$

$$\therefore F_s(\omega) = \frac{1}{T_{\max}} \sum_{n=-\infty}^{\infty} F(\omega - n\omega_s)$$

$$= 3000 \sum_{n=-\infty}^{\infty} F(\omega - 6000\pi n)$$

